SE MID PREP PRACTICE 45 MCQS ---LECTURE 03------ BY- SUBANUJAN

1. According to the lecture, what is a primary characteristic of traditional methodologies?

A) High adaptability to change

B) High level of transparency

C) Can detect errors only in the latter part of the SDLC

D) Numerous iterations

E) Low cost

Answer: C) Can detect errors only in the latter part of the SDLC

Explanation: Slide 3 explicitly lists "Can detect errors only in the latter part of the SDLC" as a disadvantage of the traditional way.

2. The modern approach to software development prioritizes which of the following?

A) Comprehensive documentation

B) Following a strict plan

C) Highest Value

D) Contract negotiation

E) Process and tools

Answer: C) Highest Value

Explanation: Slide 3 states the goals of the modern way are "Highest Value," "High Quality," and "Shortest Time."

3. The widespread adoption of Agile in SE, CS, and CSNE is due to its emphasis on:

A) Rigid planning and extensive documentation

B) Adaptability, teamwork, and iterative development

C) Contract negotiation and process adherence

D) Individual work and fixed requirements

E) Delaying integration until the end

Answer: B) Adaptability, teamwork, and iterative development

Explanation: Slide 4 directly states this as the reason for Agile's widespread adoption.

4. In which domain is Agile considered especially well-suited due to its iterative nature and focus on continuous integration?

A) Computer Science and Network Engineering (CSNE)

B) Computer Science (CS)

C) Software Engineering (SE)

D) Mechanical Engineering

E) Civil Engineering

Answer: C) Software Engineering (SE)

Explanation: Slide 6 specifies that Agile is "especially well-suited for SE" for these reasons.

5. Agile's adaptability is particularly valuable in CSNE projects because they often involve:

A) Simple user interfaces

B) Well-defined and unchanging requirements

C) Complex network infrastructure

D) Short development cycles only

E) Individual programming efforts

Answer: C) Complex network infrastructure

Explanation: Slide 8 states that CSNE projects "often involve complex network infrastructure and require flexibility."

6. Agile Software Development is an umbrella term for methods based on:

A) The IEEE Standards

B) The Waterfall Model

C) The Agile Manifesto

D) The Spiral Model

E) The V-Model

Answer: C) The Agile Manifesto

Explanation: Slide 10 defines Agile as an umbrella term for methods based on the values and principles in the Agile Manifesto.

7. According to the Agile Manifesto, what is valued more than "Comprehensive Documentation"?

A) Process and Tools

B) Contract Negotiation

C) Following a Plan

D) Working Software

E) Individual work

Answer: D) Working Software

Explanation: Slide 12's table shows the Agile value of "Working Software" over "Comprehensive Documentation."

8. Which of the following is NOT listed as an Agile methodology or practice in the slides?

A) Scrum

B) Waterfall

C) Kanban

D) Extreme Programming (XP)

E) Test Driven Development (TDD)

Answer: B) Waterfall

Explanation: Slide 13 lists Scrum, Kanban, XP, TDD, etc. Waterfall is a traditional plan-based methodology, not Agile.

9. In Scrum, the process starts by creating which artifact?

A) The Sprint Backlog

B) The Burn Down Chart

C) The Product Backlog

D) The User Manual

E) The Gantt Chart

Answer: C) The Product Backlog

Explanation: Slide 15 states: "We have to start the process by creating the Product Backlog."

10. Which of the following are the three required roles in a Scrum Framework?

A) CEO, Project Manager, Developer

B) Product Owner, ScrumMaster, Development Team

C) Architect, Tester, Business Analyst

D) User, Stakeholder, Developer

E) Product Manager, Team Lead, QA Engineer

Answer: B) Product Owner, ScrumMaster, Development Team

Explanation: Slide 16 explicitly lists these three as the required roles in the Scrum framework.

11. The Scrum role responsible for deciding which features to build and acting as the client's representative is the:

A) Development Team Lead

B) ScrumMaster

C) Project Manager

D) Product Owner

E) Stakeholder

Answer: D) Product Owner

Explanation: Slide 18 describes the Product Owner as the one who decides which features to build and is the client's representative.

12. The primary responsibility of the ScrumMaster is to:

A) Code the most complex parts of the system

B) Exert control and authority over the development team

C) Help the team understand and embrace Scrum values and practices

D) Define the product requirements and priorities

E) Accept or reject the work results

Answer: C) Help the team understand and embrace Scrum values and practices

Explanation: Slide 19 states the ScrumMaster's role is to help everyone understand and embrace Scrum.

13. How does the Development Team operate in Scrum?

A) They are managed directly by the Project Manager.

B) They follow orders from the ScrumMaster.

C) They self-organize to determine the best way to accomplish the goal.

D) They work individually on assigned tasks with no collaboration.

E) They are not cross-functional.

Answer: C) They self-organize to determine the best way to accomplish the goal.

Explanation: Slide 20 states that the development team self-organizes.

14. Which artifact is defined as a "prioritized list of work for the development team"?

A) Sprint Backlog

B) Burn Down Chart

C) Product Backlog

D) Project Charter

E) Requirement Specification Document

Answer: C) Product Backlog

Explanation: Slide 22 provides this exact definition for the Product Backlog.

15. In a well-maintained Product Backlog, where are the highest value items located?

A) At the bottom

B) Scattered throughout the list

C) In a separate document

D) At the top

E) In the middle

Answer: D) At the top

Explanation: Slide 23 states: "High Value items appear at the top of the product backlog."

16. What is a common format for expressing Product Backlog items that is understandable to both business and technical people?

A) UML Diagrams

B) Pseudocode

C) User Stories

D) Legal Contracts

E) Financial Reports

Answer: C) User Stories

Explanation: Slide 25 describes user stories as a convenient format for this purpose.

17. What is the standard template for a User Story?

A) If <condition> then <action>

B) Given <context>, when <action>, then <result>

C) As a <user role>, I want <goal>, so that <benefit>

D) The system shall <requirement>

E) <Noun> <verb> <object>

Answer: C) As a <user role>, I want <goal>, so that <benefit>

Explanation: Slide 26 shows this exact template.

18. In Scrum, a time-boxed iteration of work is called a:

A) Phase

B) Milestone

C) Release

D) Sprint

E) Cycle

Answer: D) Sprint

Explanation: Slide 32 defines a sprint as an iteration or cycle of up to a calendar month.

19. The Sprint Backlog is a subset of items taken from which larger artifact?

A) The Burn Down Chart

B) The Project Plan

C) The Product Backlog

D) The System Architecture

E) The Definition of Done

Answer: C) The Product Backlog

Explanation: Slide 32 states the sprint backlog is "a subset of Product backlog user stories."

20. What does a Burn Down Chart graphically represent?

A) Team velocity over time

B) Budget spent vs. budget remaining

C) Work left to do vs. time

D) Number of bugs found vs. fixed

E) Customer satisfaction scores

Answer: C) Work left to do vs. time

Explanation: Slide 33 defines a burn down chart as "a graphical representation of work left to do vs time."

21. On a Burn Down Chart, the outstanding work (backlog) is typically on the:

A) Horizontal axis

B) Diagonal axis

C) Vertical axis

D) Legend

E) Title

Answer: C) Vertical axis

Explanation: Slide 33 states: "The outstanding work (or backlog) is often on the vertical axis."

22. Which Scrum activity is a daily 15-minute meeting for the Development Team?

A) Sprint Planning

B) Sprint Review

C) Sprint Retrospective

D) Daily Scrum

E) Backlog Grooming

Answer: D) Daily Scrum

Explanation: Slides 39 and 40 describe the Daily Scrum as a daily, 15-minute meeting.

23. During the Daily Scrum, team members do NOT discuss:

A) What they accomplished since the last meeting

B) What they plan to work on next

C) Detailed technical solutions to problems

D) Obstacles preventing their progress

E) Their progress towards the sprint goal

Answer: C) Detailed technical solutions to problems

Explanation: The Daily Scrum (Slide 40) is for progress reporting and identifying impediments, not for detailed problem-solving, which should happen afterwards.

24. The purpose of Sprint Planning is for the Product Owner and Development Team to:

A) Conduct a retrospective on the previous sprint

B) Agree on a sprint goal for the upcoming sprint

C) Demonstrate the product increment to stakeholders

D) Hire new team members

E) Negotiate the project contract

Answer: B) Agree on a sprint goal for the upcoming sprint

Explanation: Slide 37 states that during sprint planning, they "agree on a sprint goal."

25. What is the maximum recommended duration for a Sprint?

A) 1 week

B) 2 weeks

C) A calendar month

D) 6 weeks

E) 3 months

Answer: C) A calendar month

Explanation: Slide 32 defines a sprint as "iterations or cycles of up to a calendar month."

26. Which activity provides a transparent look at the current state of the product at the end of a sprint?

A) Daily Scrum

B) Sprint Planning

C) Sprint Review

D) Sprint Retrospective

E) Backlog Refinement

Answer: C) Sprint Review

Explanation: Slide 41 states: "The sprint review provides a transparent look at the current state of the product."

27. The Sprint Retrospective is primarily for:

A) Demonstrating working software to the customer

B) Planning the features for the next release

C) Inspecting the team's process and identifying improvements

D) Writing new user stories

E) Assigning tasks to individual developers

Answer: C) Inspecting the team's process and identifying improvements

Explanation: Slide 42 describes the retrospective as a chance for the team to "analyze the way they work, identify ways to improve, and make plans to implement these improvements."

28. Who is responsible for maintaining and communicating the Product Backlog to all participants?

A) The ScrumMaster

B) The Development Team

C) The Project Manager

D) The Product Owner

E) The Stakeholders

Answer: D) The Product Owner

Explanation: Slide 18 lists this as a responsibility of the Product Owner.

29. The ScrumMaster's role differs from a traditional Project Manager because the ScrumMaster:

A) Has authority to exert control over the team

B) Is responsible for the project's budget

C) Has no authority to exert control over the team

D) Writes the code for the team

E) Defines the product vision

Answer: C) Has no authority to exert control over the team

Explanation: Slide 19 explicitly states: "The ScrumMaster has no authority to exert control over the team."

30. A key characteristic of the Development Team is that it should be:

A) Composed of specialists only

B) Managed by a functional manager

C) Cross-Functional

D) Large in size (10+ people)

E) Geographically dispersed

Answer: C) Cross-Functional

Explanation: Slide 20 lists "Cross Functional" as a characteristic of the Development Team.

31. Which of the following is a way to store a Product Backlog?

A) As a collection of index cards on a wall

B) Only in a specialized Agile tool

C) In a legally binding contract

D) In the developer's personal notebook

E) As a complete Gantt chart

Answer: A) As a collection of index cards on a wall

Explanation: Slide 22 lists this as one of the many possible ways to store the backlog.

32. The Product Backlog is best described as:

A) A fixed and unchanging list

B) A constantly evolving artifact

C) A document only for developers

D) A contract with the client

E) A list of bugs only

Answer: B) A constantly evolving artifact

Explanation: Slide 23 states: "The Product backlog is a constantly evolving artifact."

33. Which user story is better written according to the template and principles?

A) The system shall allow users to login.

B) Implement a database to store orders.

C) As a repeat customer, I want to access old orders so that I can quickly purchase the same order again.

D) User profile page must be updated.

E) Make the website faster.

Answer: C) As a repeat customer, I want to access old orders so that I can quickly purchase the same order again.

Explanation: This follows the "As a... I want... so that..." template (Slide 26) and clearly defines the user, goal, and value, unlike the other options. This comparison is also the subject of Activity 02 on Slide 30.

34. What is the primary input for creating a Sprint Backlog?

A) The CEO's vision statement

B) The prioritized Product Backlog

C) The company's annual report

D) The ScrumMaster's personal to-do list

E) The latest technology trends

Answer: B) The prioritized Product Backlog

Explanation: The Sprint Backlog is created by selecting the top-priority items from the Product Backlog (Slide 32).

35. A burn down chart is useful for:

A) Writing user stories

B) Predicting when all work will be completed

C) Hiring new team members

D) Designing the system architecture

E) Negotiating salaries

Answer: B) Predicting when all work will be completed

Explanation: Slide 33 states that the burn down chart "is useful for predicting when all of the work will be completed."

36. During which activity does the team "inspect and adapt" the product itself?

A) Sprint Retrospective

B) Daily Scrum

C) Sprint Review

D) Sprint Planning

E) Team Lunch

Answer: C) Sprint Review

Explanation: Slide 41 states the sprint review is for inspecting the product increment and adapting the Product Backlog based on feedback.

37. During which activity does the team "inspect and adapt" their process?

A) Sprint Review

B) Daily Scrum

C) Sprint Planning

D) Sprint Retrospective

E) Code Review

Answer: D) Sprint Retrospective

Explanation: Slide 42 describes the retrospective as the activity where the team examines their way of working and identifies improvements.

38. The statement "Responding to Change over Following a Plan" is one of the:

A) 12 Principles of Agile

B) 4 Key Values of Agile

C) 3 Scrum Roles

D) 5 Scrum Activities

E) 10 Scrum Artifacts

Answer: B) 4 Key Values of Agile

Explanation: Slide 11 lists the 4 values, and Slide 12 shows "Responding to Change" as one of them.

39. Which role is considered part of the broader project environment but is NOT a required role in the core Scrum framework?

A) Product Owner

B) ScrumMaster

C) Development Team Member

D) User/Stakeholder

E) This is a trick question; all are required.

Answer: D) User/Stakeholder

Explanation: Slide 20 lists User/Stakeholders separately from the three required Scrum roles (PO, SM, Dev Team) mentioned on Slide 16.

40. The main goal of Agile methodologies is to deliver:

A) Extensive documentation first

B) The highest value to the customer in the shortest time

C) A perfect plan before any code is written

D) A fixed set of features as defined in an initial contract

E) Software only at the very end of the project

Answer: B) The highest value to the customer in the shortest time

Explanation: This encapsulates the modern way's goals from Slide 3 (Highest Value, Shortest Time) and is the overarching theme of Agile.

41. Which of the following are explicitly listed as disadvantages of the "Traditional Way" of development? (Select all that apply)

A) High Cost

B) Lack of transparency

C) Too many meetings

D) Changes are not acceptable

E) Can detect errors only in the latter part of the SDLC

Answer: A) High Cost, B) Lack of transparency, D) Changes are not acceptable, E) Can detect errors only in the latter part of the SDLC

Explanation: Slide 3 lists all of these except "Too many meetings" as characteristics of the traditional methodology.

42. The Agile Manifesto consists of which of the following? (Select all that apply)

A) 4 Key Values

B) 12 Principles

C) 3 Core Roles

D) 5 Ceremonies

E) A formal proclamation

Answer: A) 4 Key Values, B) 12 Principles, E) A formal proclamation

Explanation: Slide 10 states the Agile Manifesto is a formal proclamation of 4 key values and 12 principles. The roles and ceremonies are specific to frameworks like Scrum, not the overarching Manifesto.

43. Which of the following statements correctly describe the Product Owner's responsibilities? (Select all that apply)

A) They shield the team from external interferences.

B) They decide on release date and content.

C) They are the empowered central point of product leadership.

D) They help the team understand Scrum practices.

E) They accept or reject work results.

Answer: B) They decide on release date and content, C) They are the empowered central point of product leadership, E) They accept or reject work results.

Explanation: These are all direct responsibilities of the Product Owner listed on Slide 18. Shielding the team (A) and helping them understand Scrum (D) are responsibilities of the ScrumMaster (Slide 19).

44. What are the three required artifacts in Scrum according to the lecture slides?

(Select three)

A) User Stories

B) Product Backlog

C) Burn Down Charts

D) Sprint Backlog

E) UML Diagrams

Answer: B) Product Backlog, C) Burn Down Charts, D) Sprint Backlog

Explanation: Slide 21 explicitly lists these three as the "SCRUM Artifacts." User Stories are a format for backlog items, and UML Diagrams are a possible tool but not a required Scrum artifact.

45. Consider the following statements about a Sprint. Which of them are TRUE?

(Select all that apply)

A) Its duration can be up to one calendar month.

B) No changes can be made that would alter the sprint goal.

C) The scope is negotiated and can change daily.

D) It has an agreed-upon definition of "Done."

E) It begins with a Sprint Retrospective meeting.

Answer: A) Its duration can be up to one calendar month, B) No changes can be made that would alter the sprint goal, D) It has an agreed-upon definition of "Done."

Explanation: A and B are confirmed by Slide 38 ("Short duration," "No goal-altering changes"). D is also from Slide 38 ("Agreed-upon definition of done"). The scope is fixed

once the sprint starts (C is false), and a sprint begins with Sprint Planning, not a Retrospective (E is false).